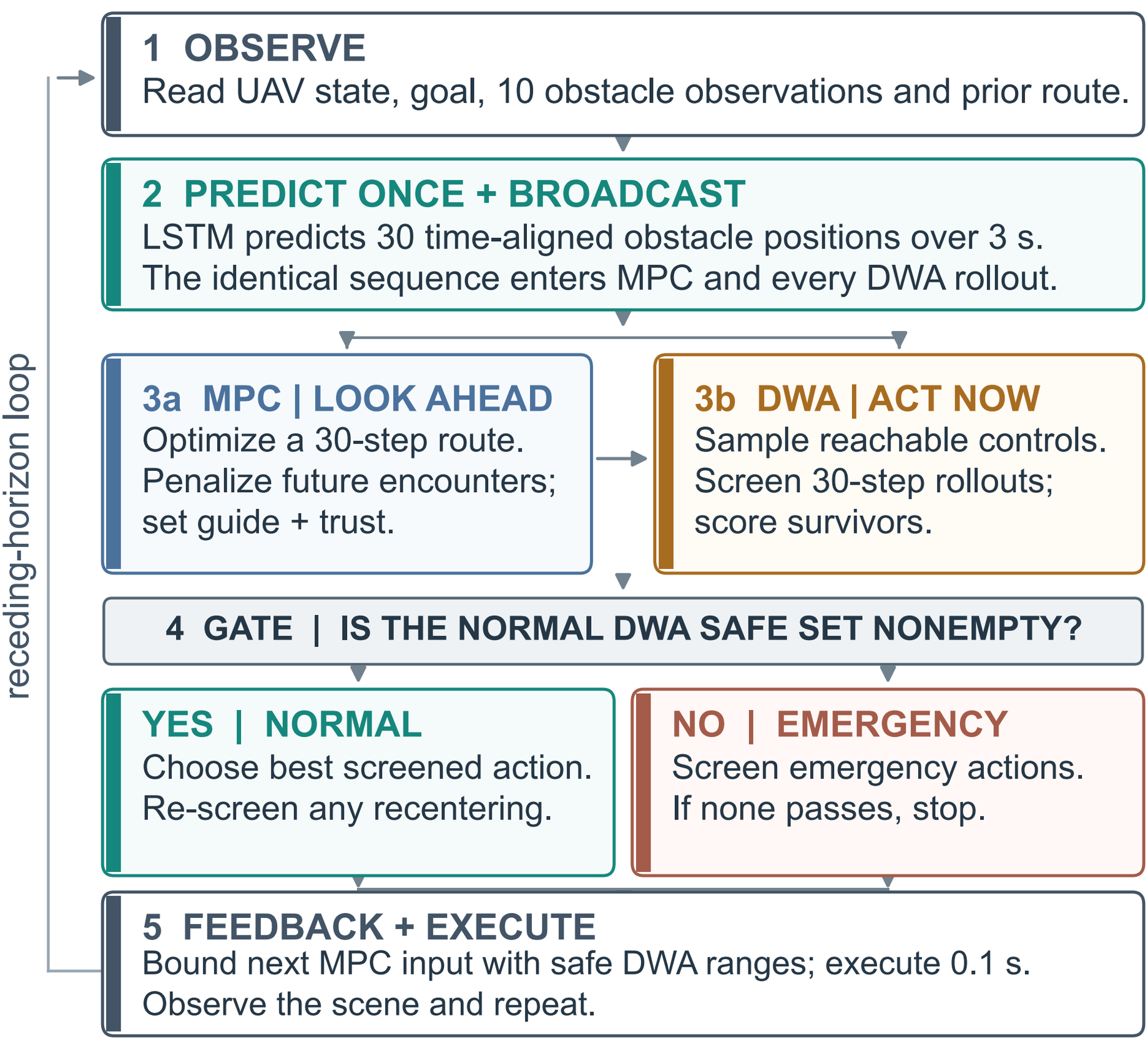
SHARED-FORECAST MPC--DWA | ONE UPDATE
ONE FUTURE | TWO TIME SCALES | SCREENED EXECUTION
1 OBSERVE
Read UAV state, goal, 10 obstacle observations and prior route.
2 PREDICT ONCE + BROADCAST
LSTM predicts 30 time-aligned obstacle positions over 3 s.
The identical sequence enters MPC and every DWA rollout.
3a MPC | LOOK AHEAD
Optimize a 30-step route.
Penalize future encounters;
set guide + trust.
3b DWA | ACT NOW
Sample reachable controls.
Screen 30-step rollouts;
score survivors.
4 GATE | IS THE NORMAL DWA SAFE SET NONEMPTY?
YES | NORMAL
Choose best screened action.
Re-screen any recentering.
NO | EMERGENCY
Screen emergency actions.
If none passes, stop.
5 FEEDBACK + EXECUTE
Bound next MPC input with safe DWA ranges; execute 0.1 s.
Observe the scene and repeat.
receding-horizon loop